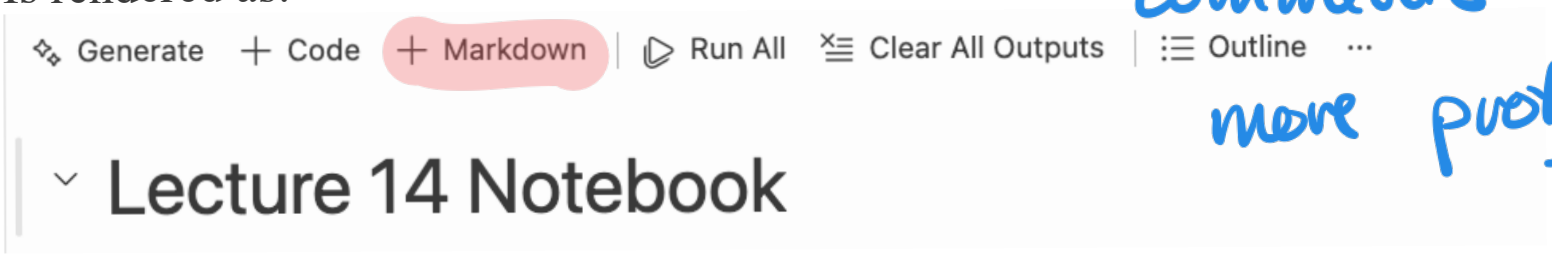
CSC 251 Lecture 14 – Markdown, Correlation, Linear Models

Markdown in Jupyter Notebooks

Easy-to-use markup language used to add formatting to plain text without formal HTML tags.
Example 1a: In a markdown cell in your Notebook:

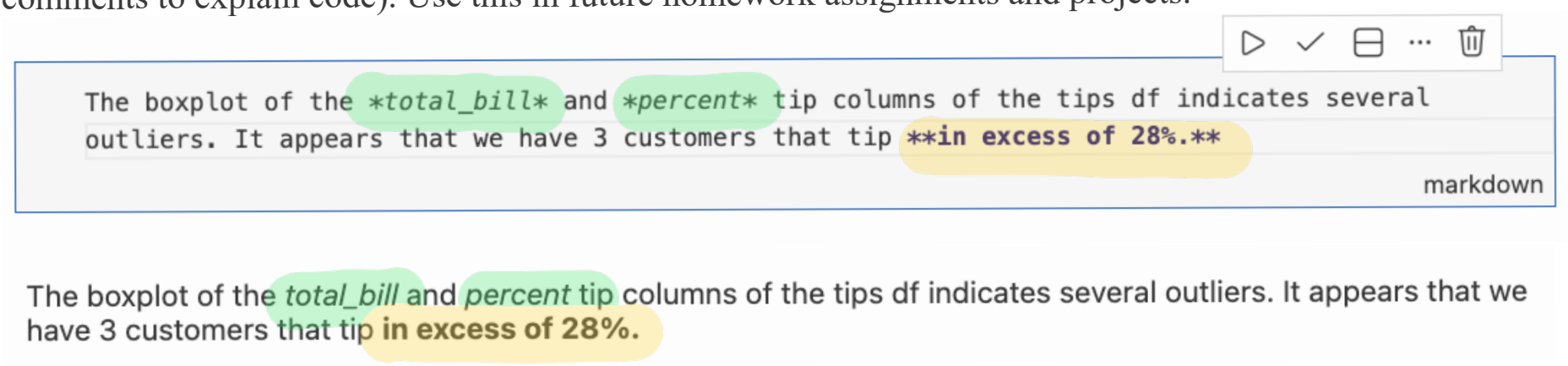
```
# Lecture 14 Notebook
```

Is rendered as:



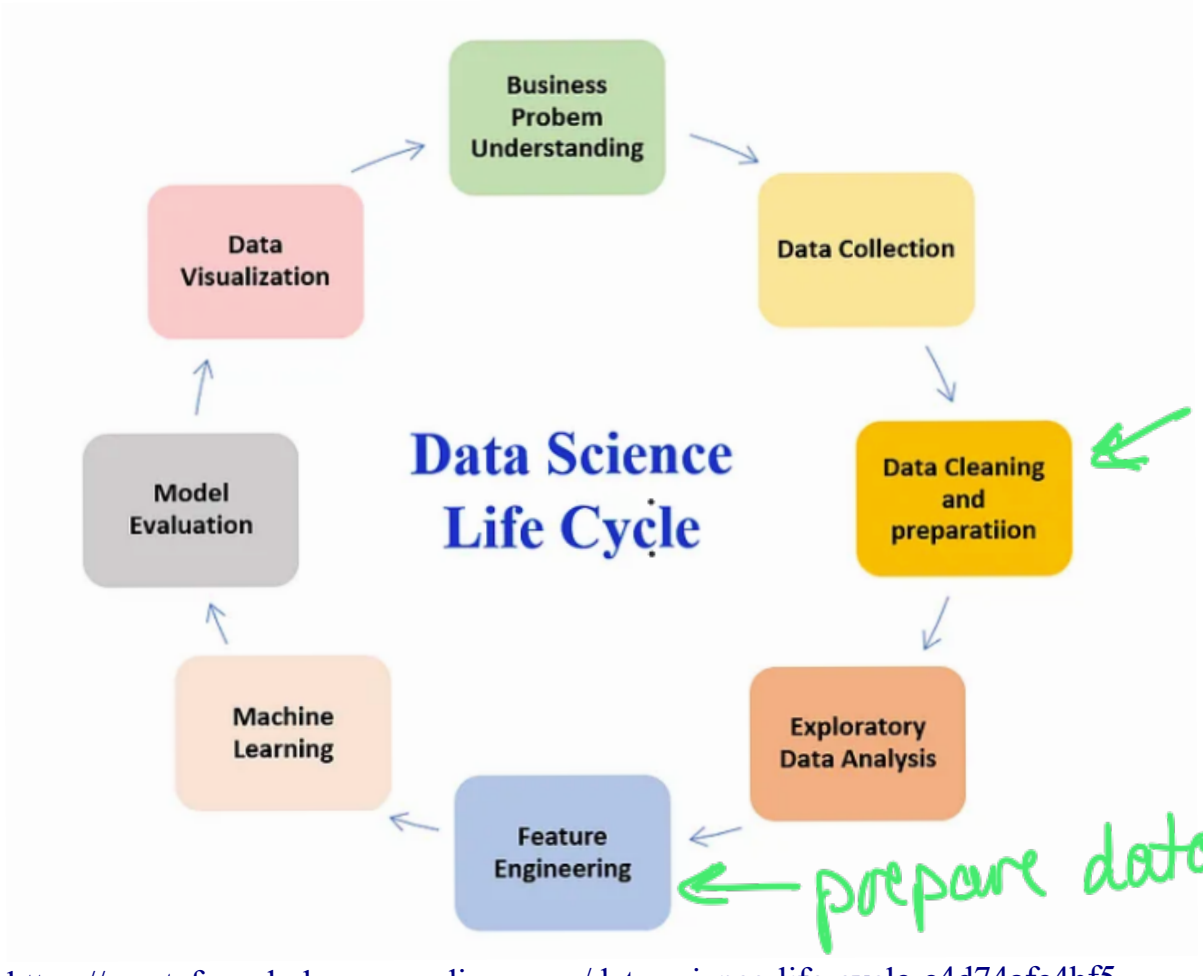
Replace code comments with markdown comments - makes it more professional!

Example 1b: Use markdown cells for documentation and to provide a running narrative of your notebook (along with comments to explain code). Use this in future homework assignments and projects.



References:
https://www.markdownguide.org/basic-syntax/
https://jupyter-notebook.readthedocs.io/en/stable/examples/Notebook/Working%20With%20Markdown%20Cells.html

The Data Science Life Cycle

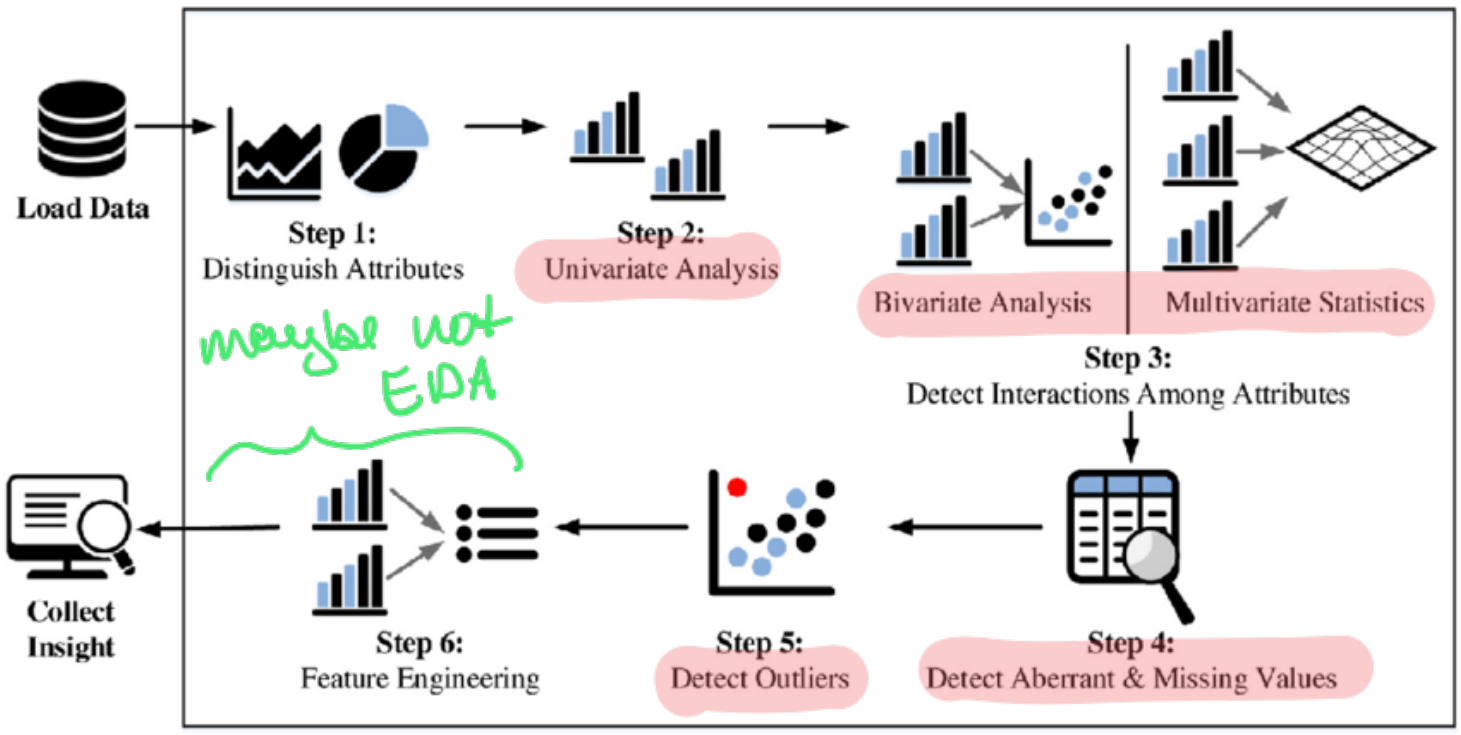


can be time-consuming

prepare data for ML

https://mustafaserdarkonca.medium.com/data-science-life-cycle-e4d74afe4bf5

Exploratory Data Analysis



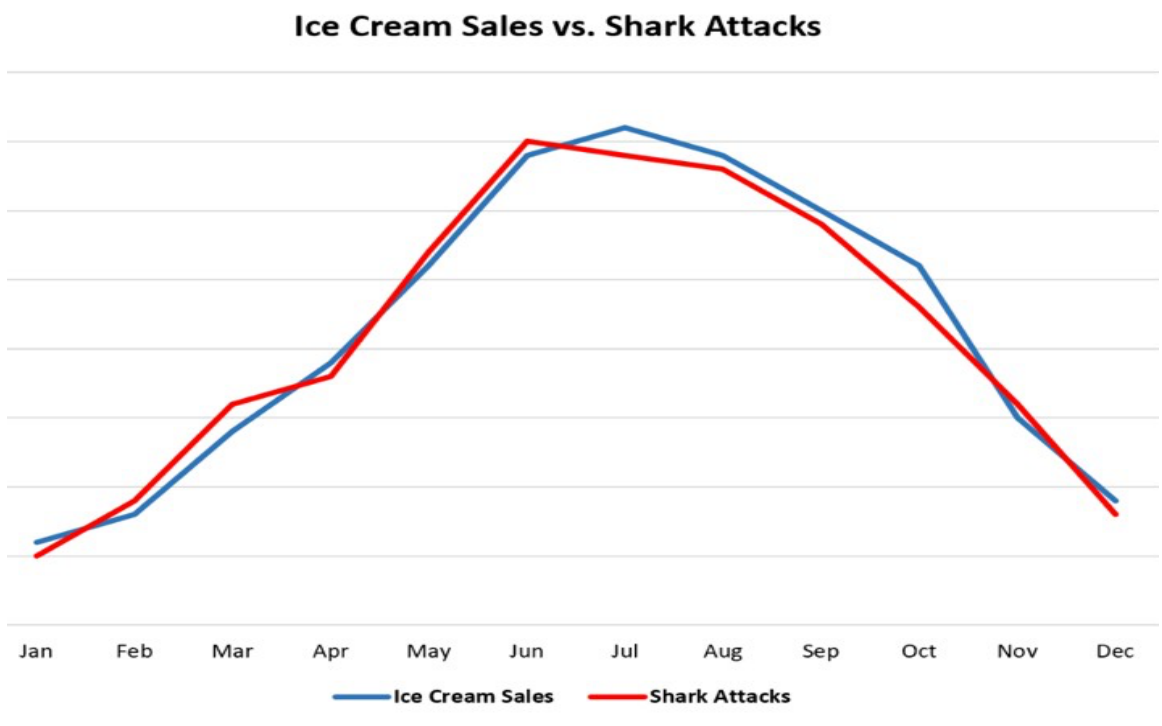
https://www.researchgate.net/figure/The-fundamental-steps-of-the-exploratory-data-analysis-process_fig3_329930775



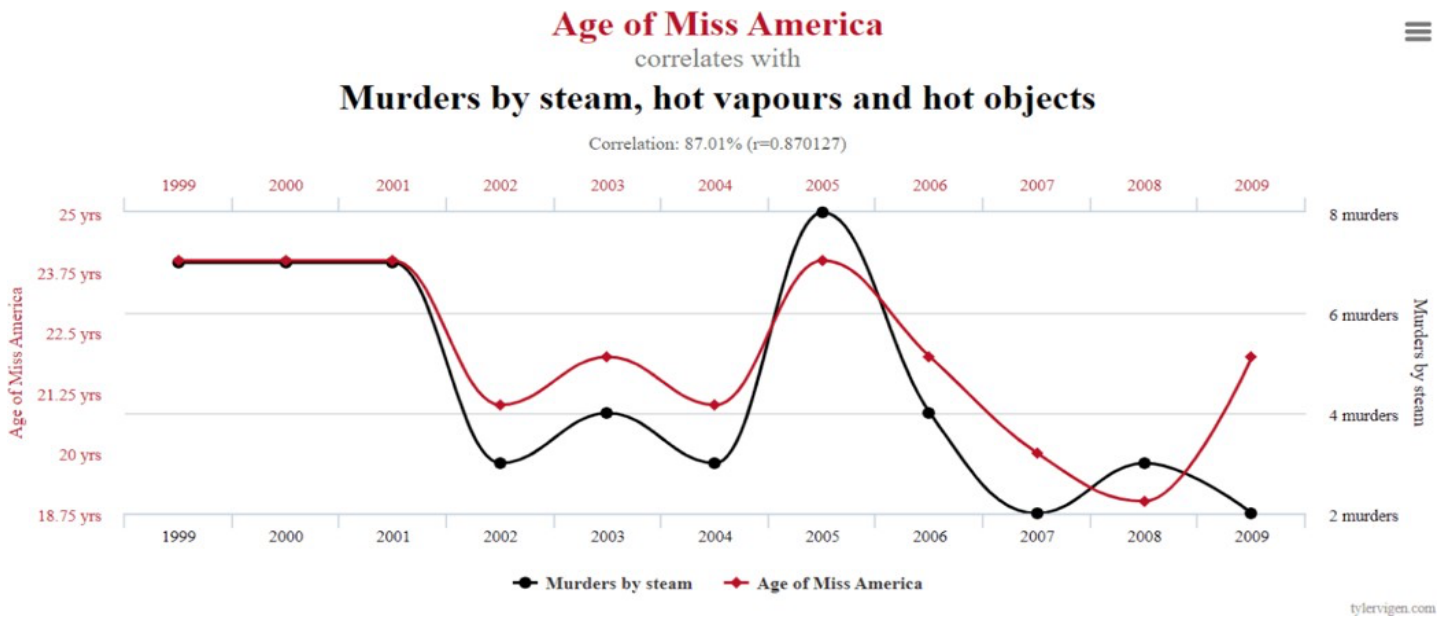
https://www.excelr.com/exploratory-data-analysis-in-data-science

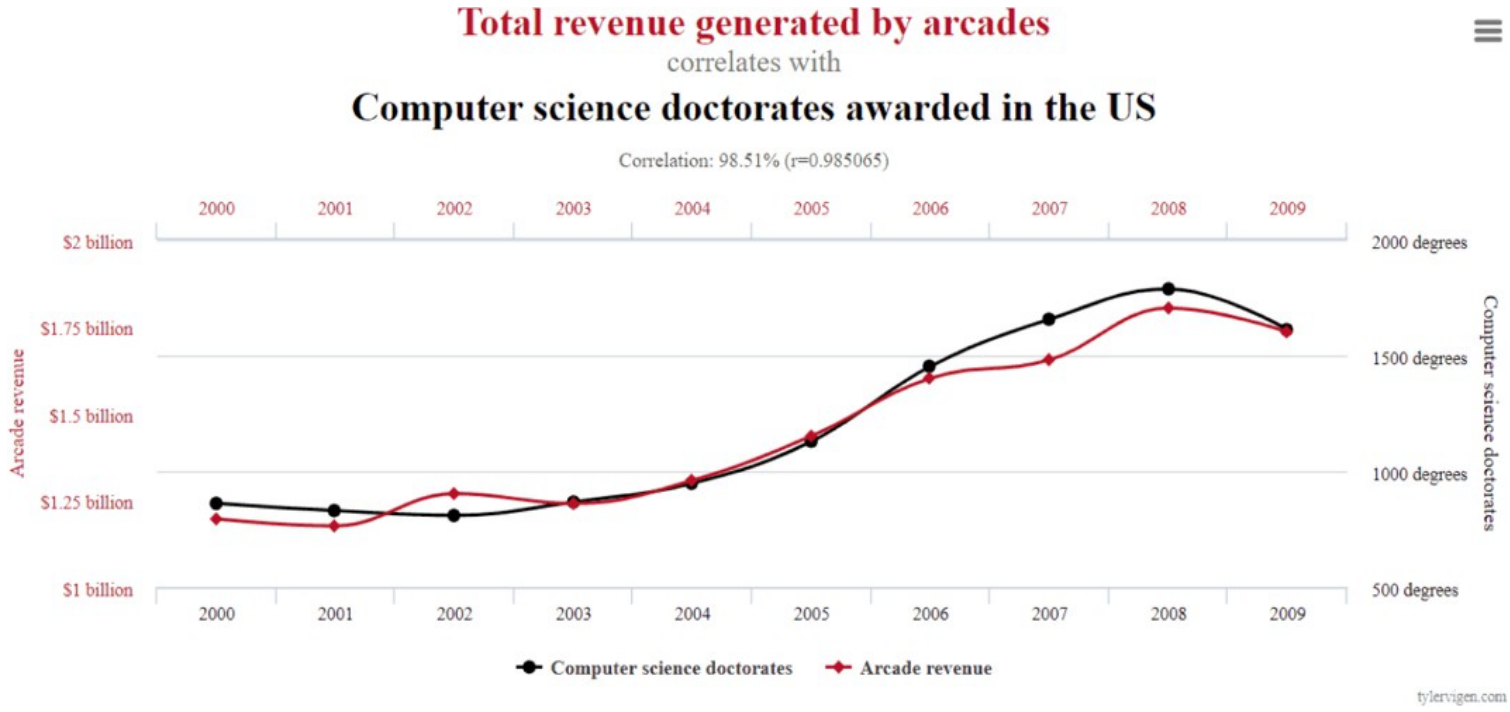
Causation versus Correlation

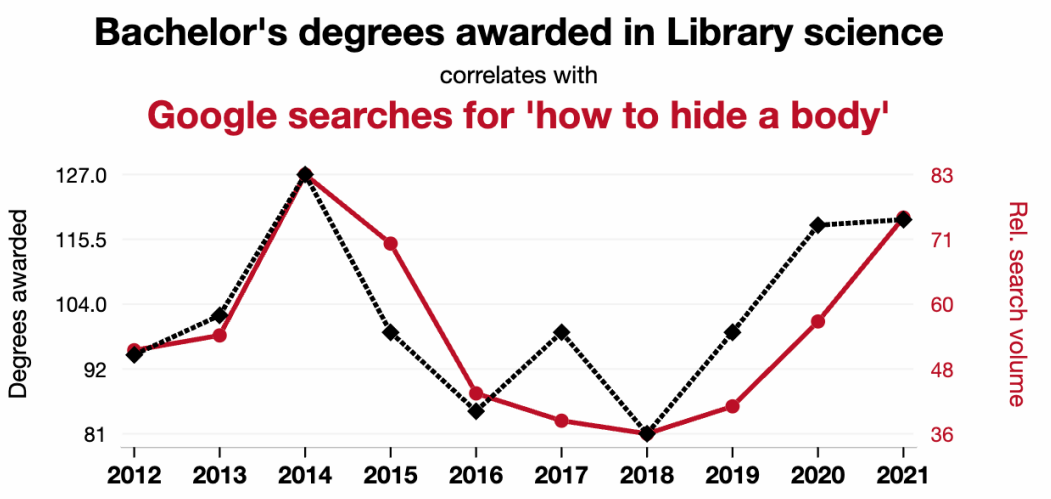
During Exploratory Data Analysis, we investigate possible relationships between variables.
"Does the change in one variable correlate to a change in another variable?"



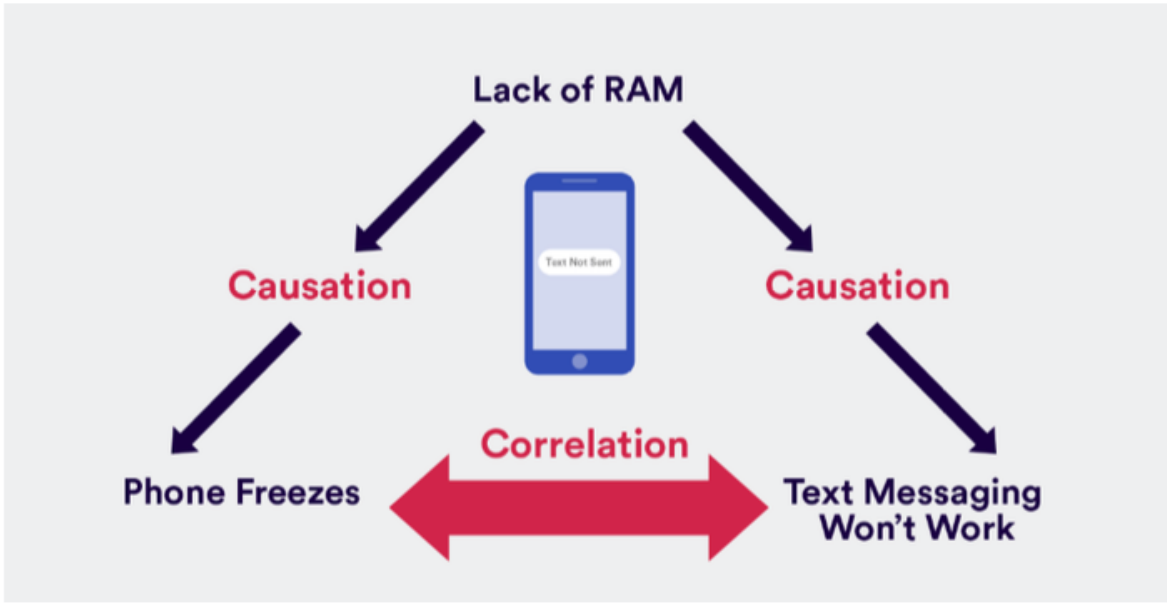
both related to hot weather and beach vacations







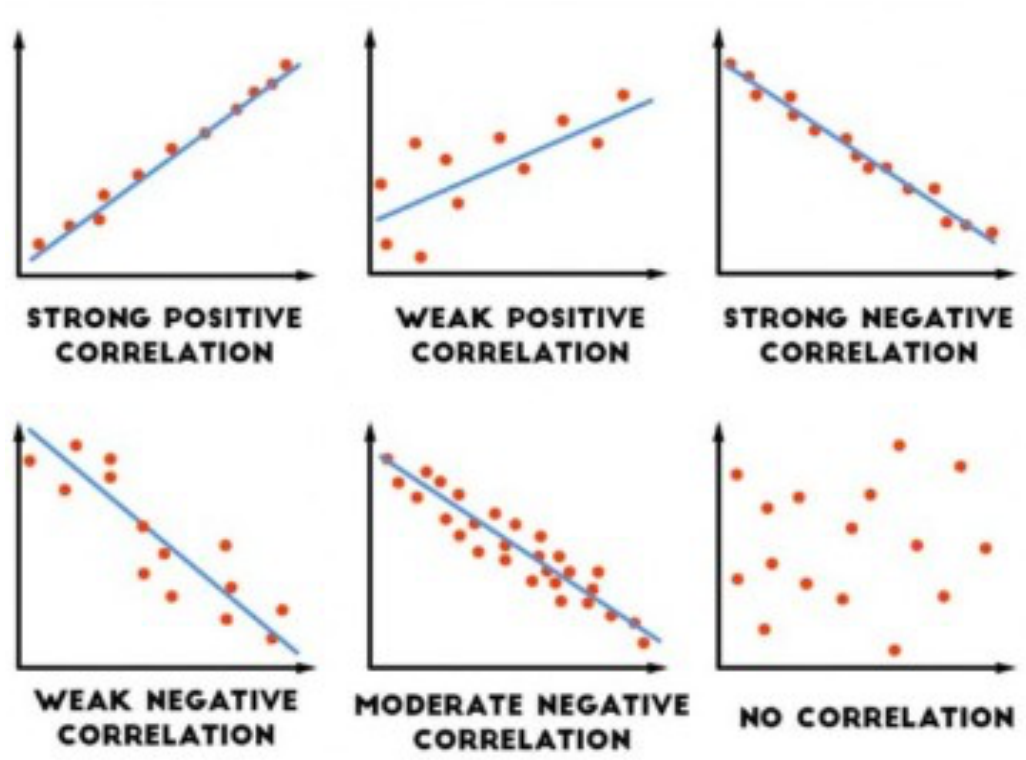
While correlation doesn’t imply causation, if there were causation, there would be a correlation.



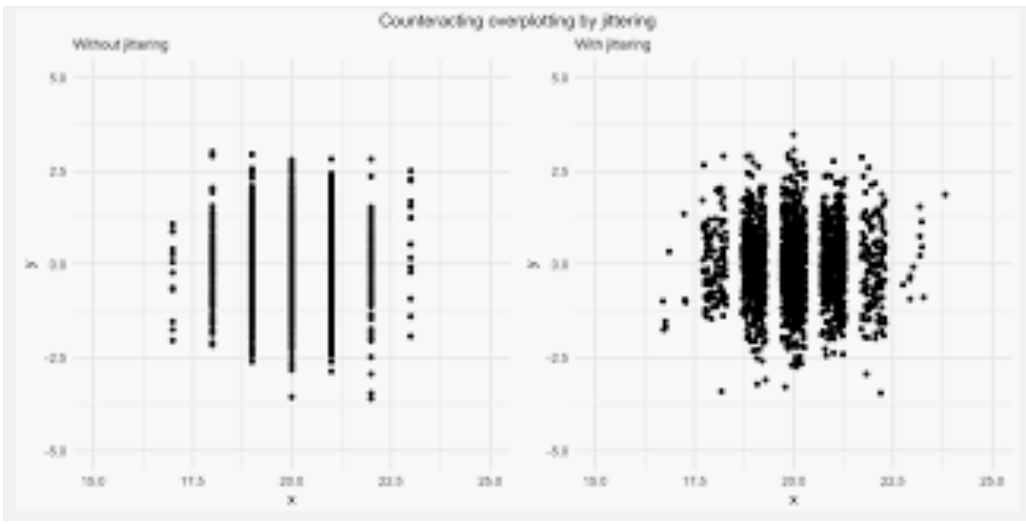
Identifying Correlation

1) Scatter plot for diagrammatic approach

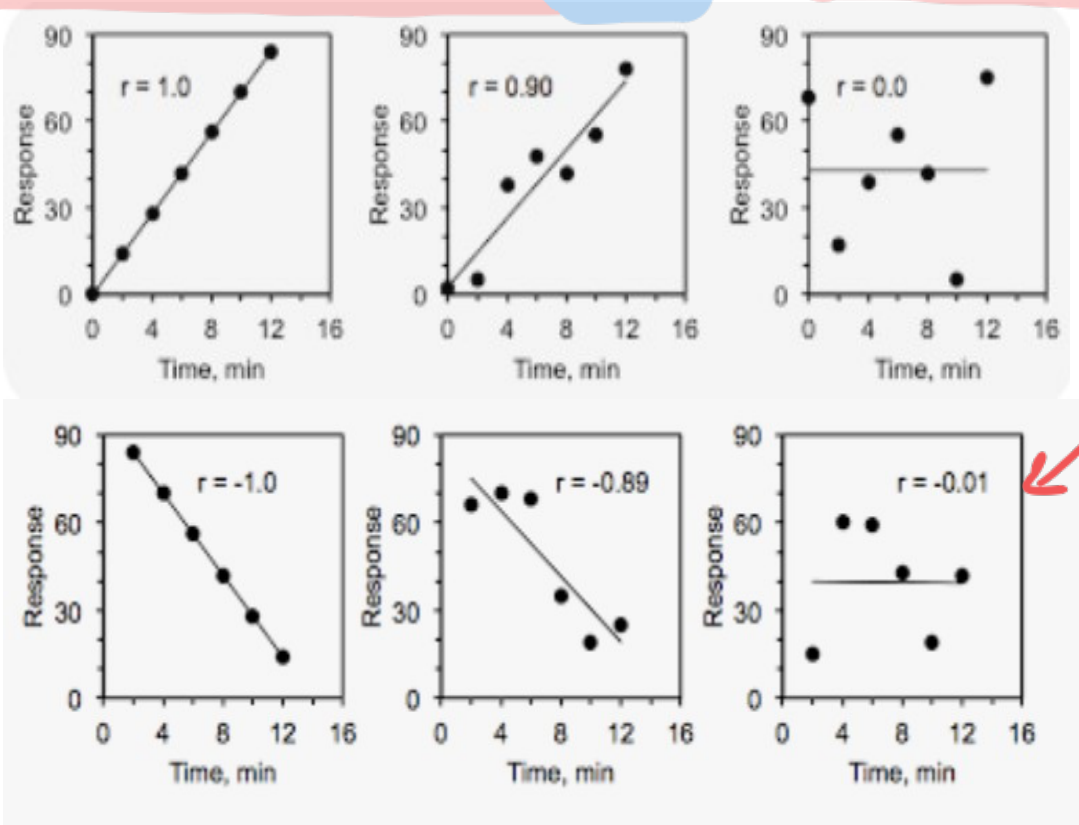
Direction and intensity of correlation:



Beware of overplotting – values may overlap and misrepresent the data. Try "jittering" the data.



2) Pearson's Correlation Coefficient (r-value) `-1 <= r <= 1` where r = 0 indicates no correlation



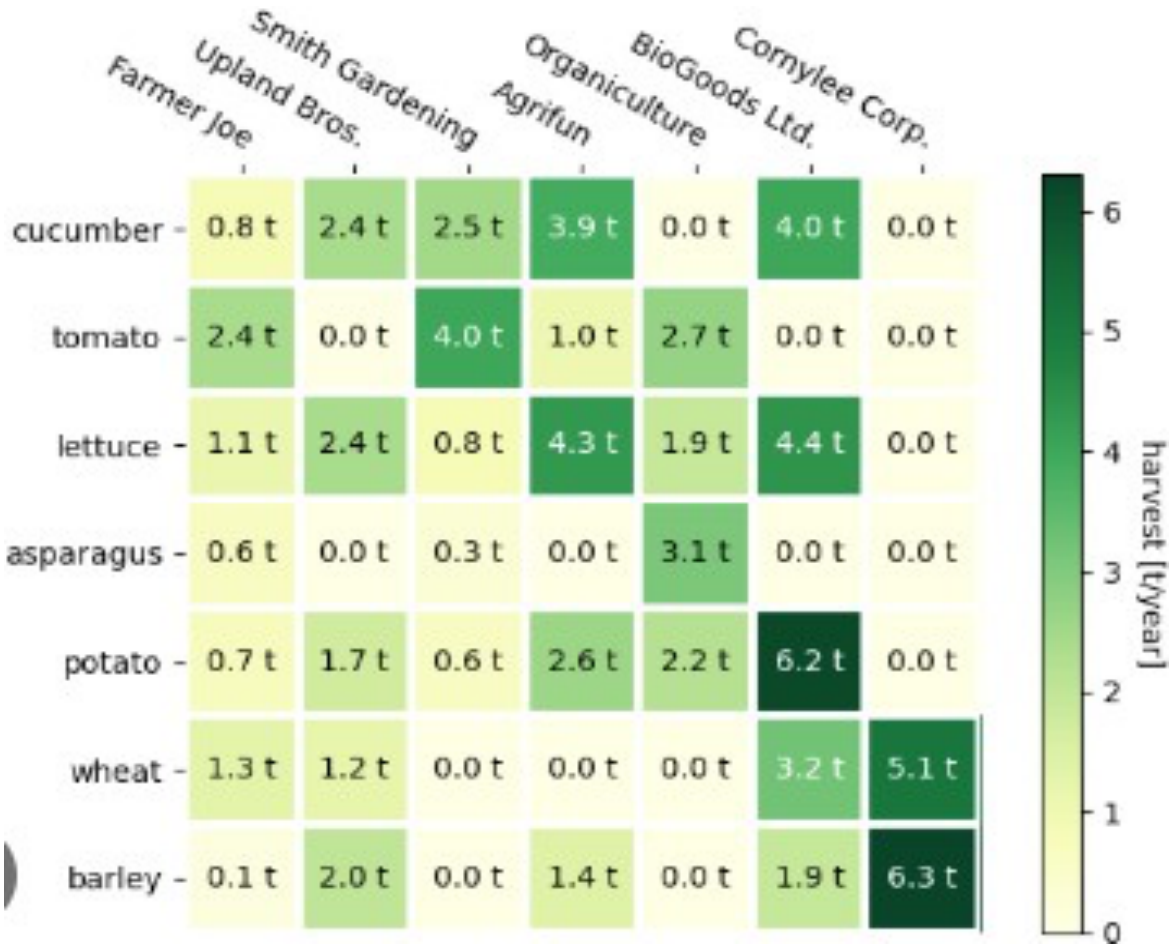
Just because you can fit a line doesn't mean you should!

Only for linear relationships, valid for dichotomous variables (categorical variables with 2 values)
To be sign-agnostic, sometimes an R-squared value is reported.

What value of r determines a strong enough correlation? .9? .6? .4?

Depends upon data, human-related data is messy!

3) Heat map – graphical representation with shades to indicate strength of correlation:



Example 2: Consider the **mpg** DataFrame:

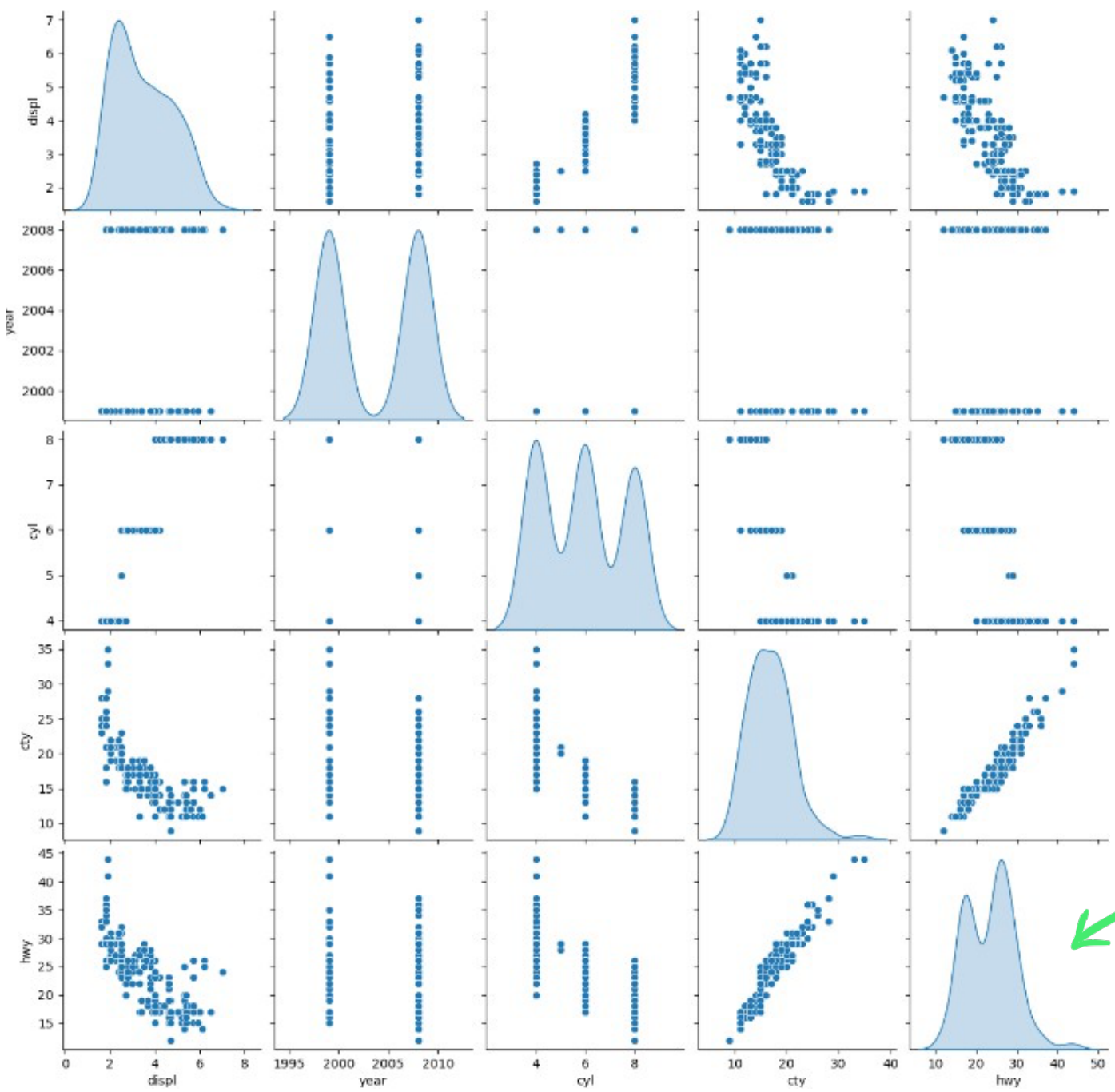
0	manufacturer	model	displ	year	cyl	trans	drv	cty	hwy	fl	class
1	audi	a4	1.8	1999	4	auto(l5)	f	18	29	p	compact
2	audi	a4	1.8	1999	4	manual(m5)	f	21	29	p	compact
3	audi	a4	2.0	2008	4	manual(m6)	f	20	31	p	compact
4	audi	a4	2.0	2008	4	auto(av)	f	21	30	p	compact
5	audi	a4	2.8	1999	6	auto(l5)	f	16	26	p	compact

(See Lecture 14 Notebook)

Create a pairplot to identify possible correlations picking non-string (object) variables:

less customizable, but less code

```
import seaborn as sns
sns.pairplot(data = mpg, y_vars=['displ', 'year', 'cyl', 'cty', 'hwy'],
             x_vars=['displ', 'year', 'cyl', 'cty', 'hwy'], diag_kind='kde')
```



univariate density plots on diagonal

Check the correlation values:

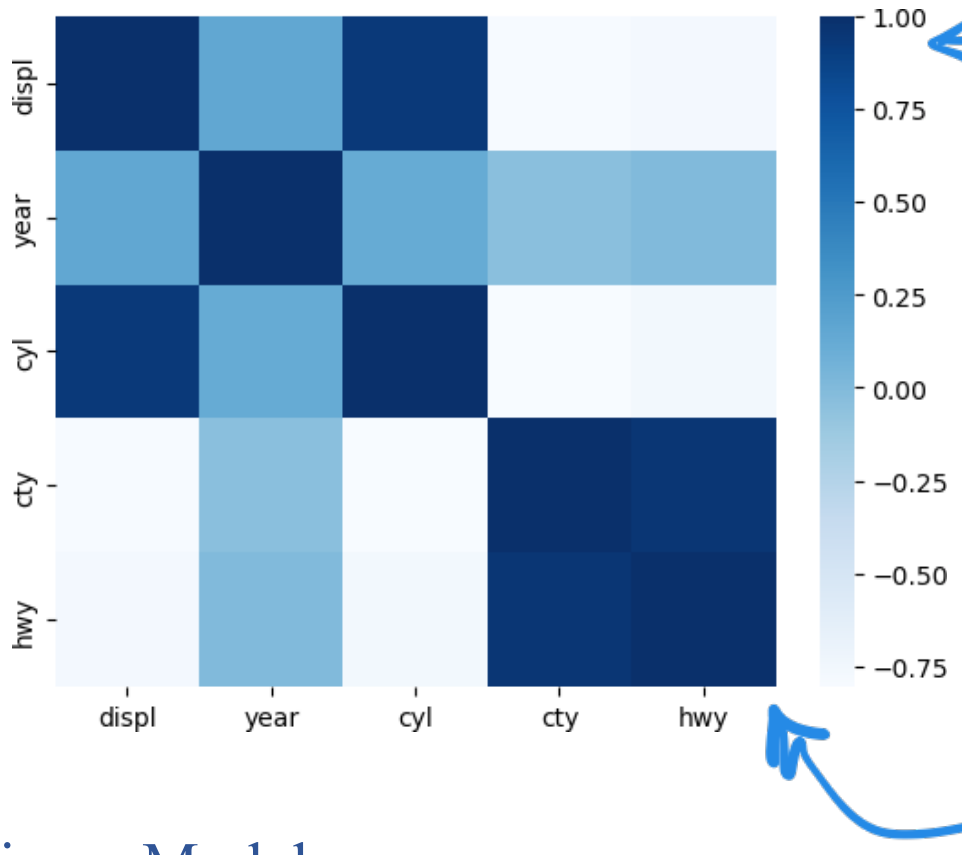
```
mpg[['displ', 'year', 'cyl', 'cty', 'hwy']].corr()
```

	displ	year	cyl	cty	hwy
displ	1.000000	0.147843	0.930227	-0.798524	-0.766020
year	0.147843	1.000000	0.122245	-0.037232	0.002158
cyl	0.930227	0.122245	1.000000	-0.805771	-0.761912
cty	-0.798524	-0.037232	-0.805771	1.000000	0.955916
hwy	-0.766020	0.002158	-0.761912	0.955916	1.000000

How high does r need to be? Here, maybe > 0.7

Create a heatmap:

```
mpg2 = mpg[['displ', 'year', 'cyl', 'cty', 'hwy']]
sns.heatmap(data=mpg2.corr(), cmap = 'Blues')
```

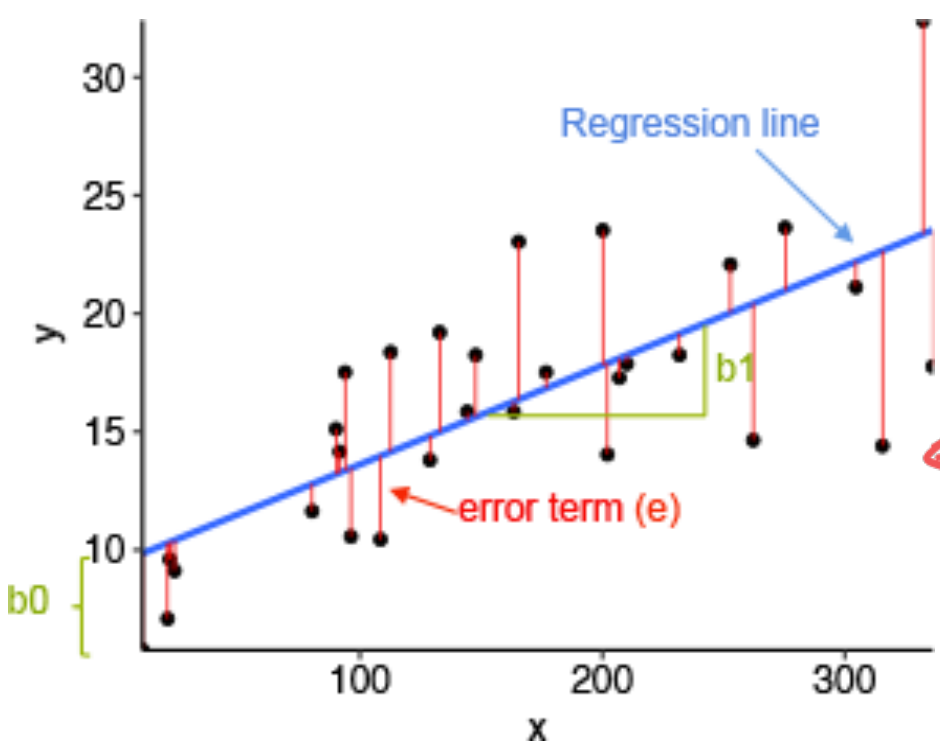


dark blue indicates strongest correlation

Linear Models

For variables that appear to have some correlation, we want to create a predictive model. The most popular model is a **linear regression model** that finds the best fit linear relationship between a response (dependent) variable and a predictor (independent) variable. Also called **ordinary least squares** fit or best fit line and comes in both a simple or multiple variety depending on the number of independent variables used.

Given a set of data points, we can find the line that minimizes the "residuals" – the distances from the actual values to the predicted values.



minimize residuals

pts weighted equally

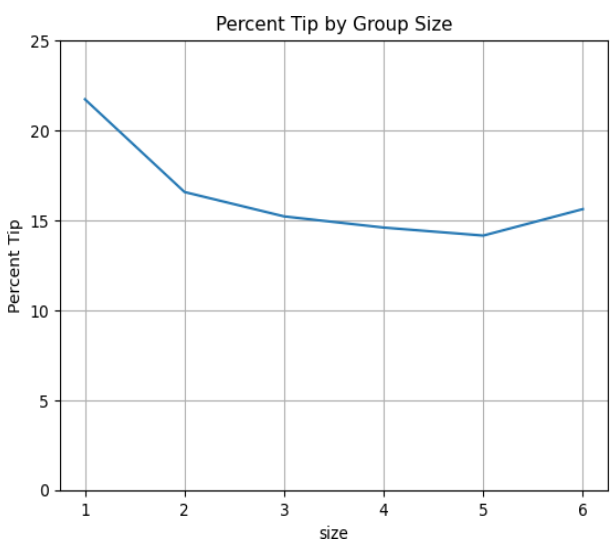
Python packages for modeling:

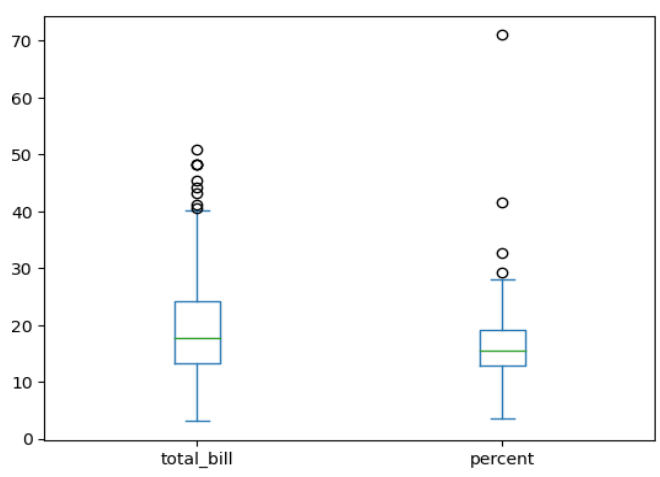
1) statsmodels – "classical" (non-Bayesian) statistical models, ANOVA methods, time series processes
2) scikit-learn – general purpose machine learning toolkit

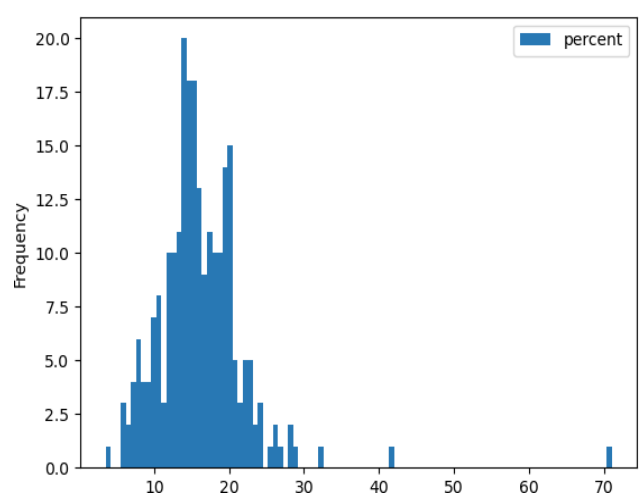
Example 3: Let's take another look at the **tips** DataFrame:

	total_bill	tip	smoker	day	time	size	percent
0	16.99	1.01	No	Sun	Dinner	2	5.944673
1	10.34	1.66	No	Sun	Dinner	3	16.054159
2	21.01	3.50	No	Sun	Dinner	3	16.658734
3	23.68	3.31	No	Sun	Dinner	2	13.978041
4	24.59	3.61	No	Sun	Dinner	4	14.680765
...	...	...	...	...	...	...	...

Recall some of the visualizations we created:

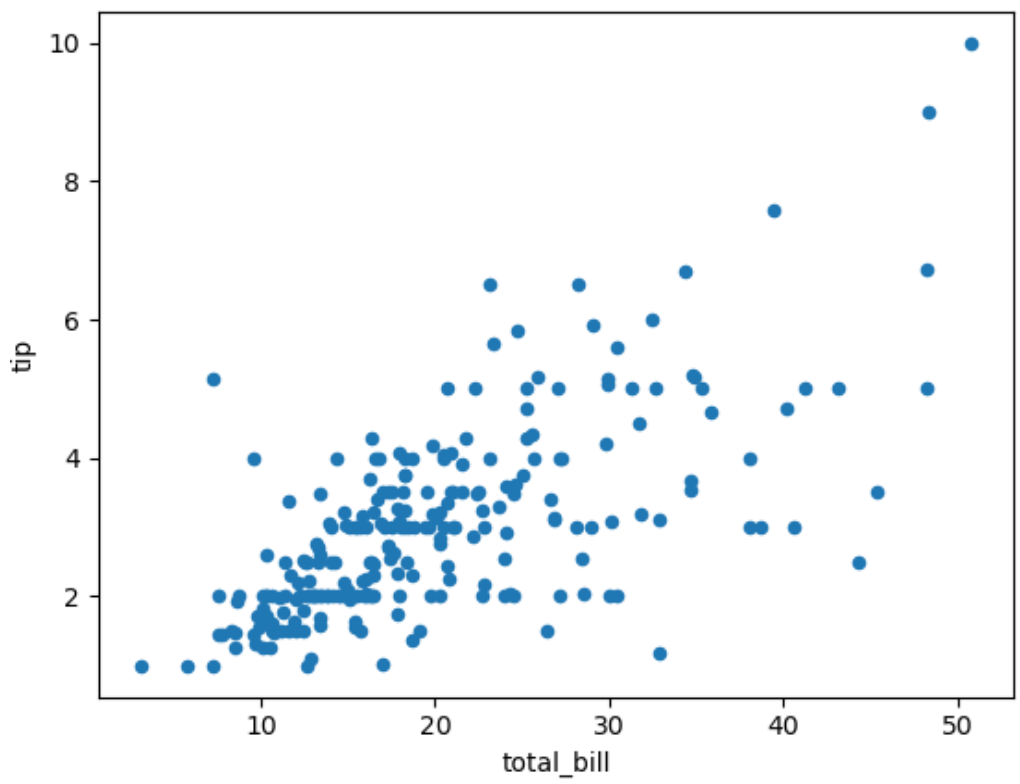






Create a scatter plot to look for correlation:

```
tips.plot.scatter(x = 'total_bill', y = 'tip')
```



Some correlation! Not perfect since human judgement is involved!

How strong is the correlation? To use the Ordinary Least Squares (OLS) method from statsmodels:

```
import statsmodels.formula.api as smf
results = smf.ols(formula = 'tip ~ total_bill', data = tips).fit()
results.summary()
```

y x tilda

What's important?

OLS Regression Results

Dep. Variable:	tip	R-squared:	0.457
Model:	OLS	Adj. R-squared:	0.454
Method:	Least Squares	F-statistic:	203.4
Date:	Wed, 01 Nov 2023	Prob (F-statistic):	6.69e-34
Time:	16:33:11	Log-Likelihood:	-350.54
No. Observations:	244	AIC:	705.1
Df Residuals:	242	BIC:	712.1
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P>|t|	[0.025	0.975]
Intercept	0.9203	0.160	5.761	0.000	0.606	1.235
total_bill	0.1050	0.007	14.260	0.000	0.091	0.120

b m

Omnibus:	20.185	Durbin-Watson:	2.151
Prob(Omnibus):	0.000	Jarque-Bera (JB):	37.750
Skew:	0.443	Prob(JB):	6.35e-09
Kurtosis:	4.711	Cond. No.	53.0

total bill → $y = mx + b$ ← tip

always replace x and y with actual variables

tip = .105 * total_bill + .920

We can also get the r correlation coefficient:

```
tips[['tip', 'total_bill']].corr()
```

	tip	total_bill
tip	1.000000	0.675734
total_bill	0.675734	1.000000

← r-values

To find the equation of our line, y = mx + b, the intercept and slope are stored in a tuple:

```
results.params
```

returns a series

```
intercept 0.920270
total_bill0.105025
dtype:  float63
```

0 1

ALWAYS replace 'x' with the name of the predictor (independent) variable name and replace 'y' with the response (dependent) variable name from the data you are modeling!

What does the slope tell us? Intercept?

The additional amount of tip for each additional $1 spent

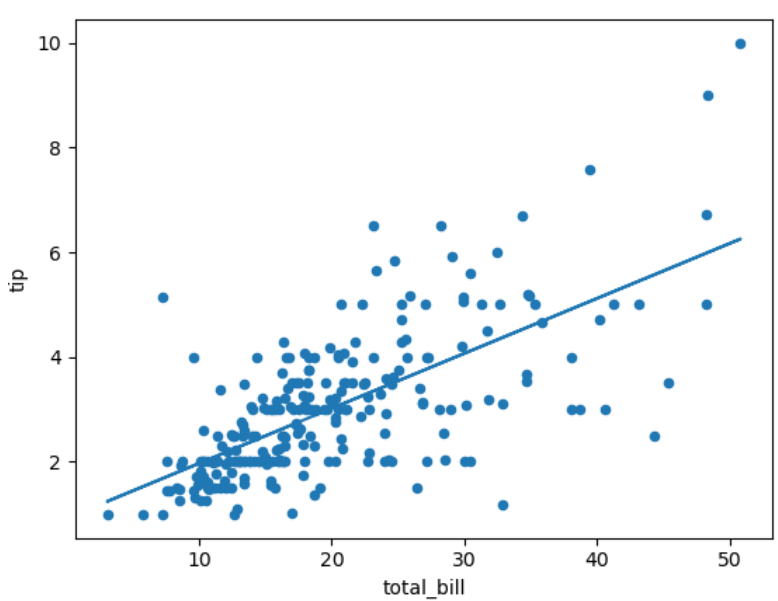
The tip for $0 bill → non-sensical

Graphing options:

1) Add a column in the DataFrame and layer the plots:

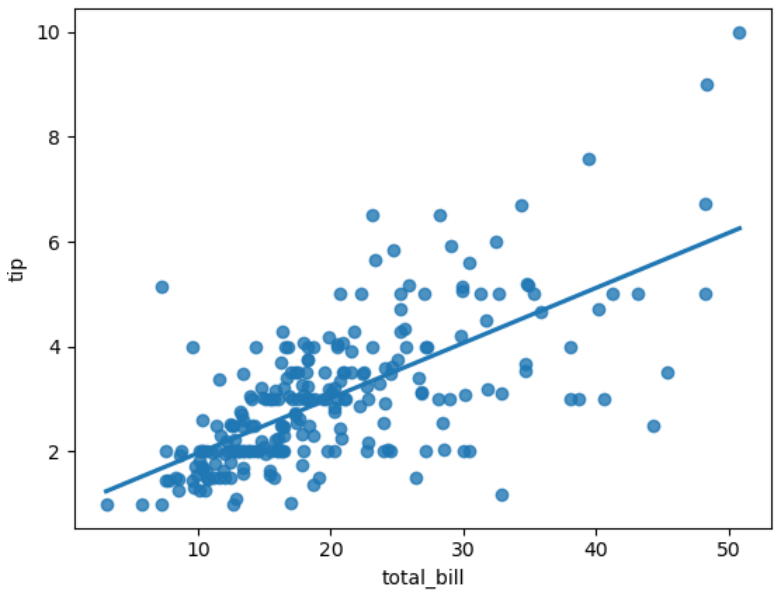
```
tips['line'] = tips['total_bill'] *results.params[1] + results.params[0]

ax = tips.plot(x='total_bill', y = 'tip', kind = 'scatter')
ax.plot('total_bill', 'line', data = tips)
```



2) Use Seaborn to easily create the graph:

```
import seaborn as sns
sns.regplot(x='total_bill', y = 'tip', ci = None, data = tips)
```



Faster and easier but no access to predicted values!